

Sparse-Autoencoder Dissection of Emotion Features in Frozen Speech Encoders

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USC ICT — Intelligent Human Perception Lab · mechanistic extension of linguistic-agnostic-ser

Abstract — Linear probes show emotion is decodable in the middle-to-late layers of frozen speech transformers, but not which internal features carry it, nor whether the signal is acoustic or lexical. We train TopK Sparse Autoencoders (SAEs) on per-frame activations of six frozen large encoders (HuBERT, WavLM, wav2vec2, w2v-BERT, MERT, Whisper) and dissect their emotion features on the lexicon-controlled CREMA-D corpus (91 actors x 12 fixed sentences x 6 emotions, 7,442 clips). A confound-controlled selection — a feature's firing must be more informative about emotion than about the spoken sentence or the speaker — isolates genuine emotion features from phonetic and speaker-driven ones. We find (i) emotion DISENTANGLES into clean monosemantic features earlier (L2-L7) than where it is best linearly DECODED (L10); (ii) all six encoders share an acoustic emotion-feature basis that generalizes across unseen speakers; and (iii) the recovered emotion signal is acoustic, not lexical. Decodability and disentanglement are distinct, and emotion here is carried by speaker-invariant acoustic features.

1. Introduction

Self-supervised speech encoders dominate speech-emotion recognition, yet were trained on ASR-style objectives that encode linguistic structure. Probing tells us WHERE emotion is readable, not which features encode it or whether the signal is acoustic or lexical. SAEs decompose dense activations into sparse, near-monosemantic features, moving the analysis from decodability to mechanism. CREMA-D's fixed lexicon lets us dissociate acoustic delivery from text by construction, so any recovered emotion signal there is necessarily acoustic.

2. Method

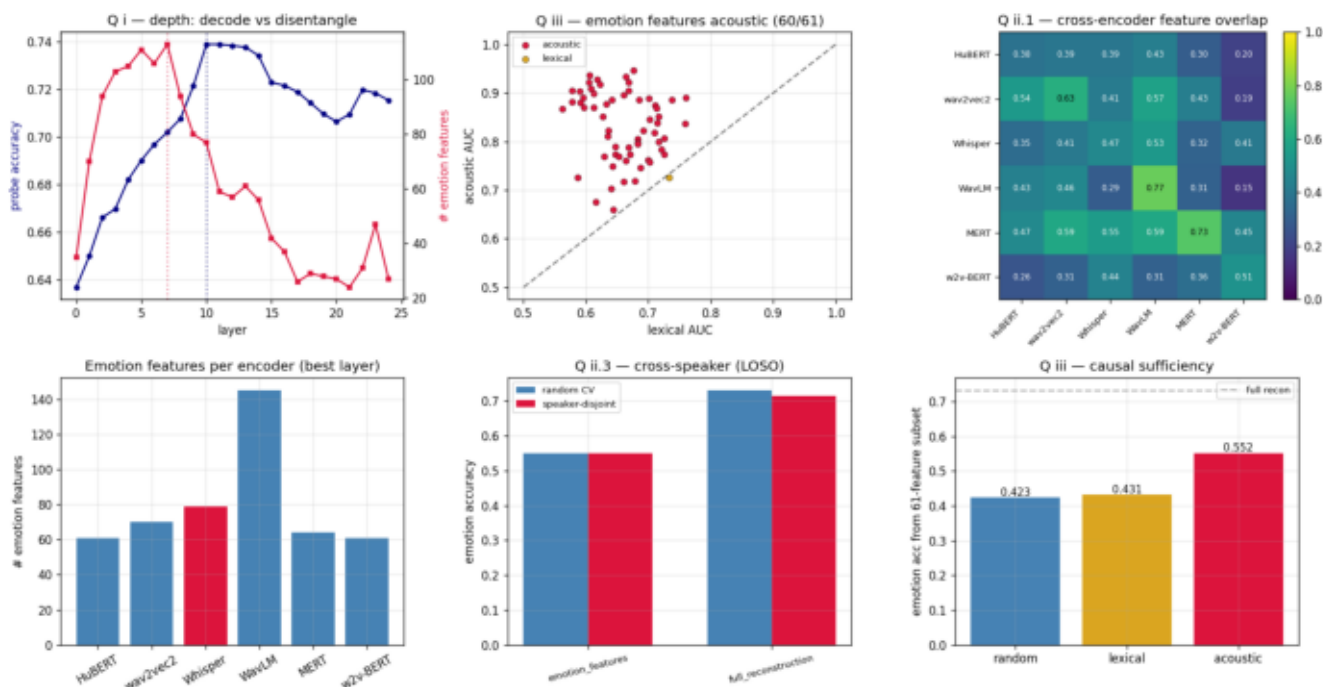
We re-extract UN-pooled per-frame hidden states from the frozen encoders (the project's cached embeddings are mean-pooled, too few to train an over-complete SAE); CREMA-D L13 yields 940,273 frames. On these we train a TopK SAE ($d_{in}=1024$, $d_{sae}=8192$, $k=32$, AuxK dead-feature revival), reaching FVU=0.070 (93% variance explained), 4/8192 dead features.

Emotion features are scored by mutual information (MI) between a feature's firing and the emotion label, thresholded at 5 sigma over a label-permutation null. A feature qualifies only if MI(emotion) exceeds MI(sentence) AND MI(actor) and it is monosemantic (≥ 0.5 activation mass on one emotion). Acoustic-vs-lexical labeling uses cross-validated ROC-AUC of a logistic classifier predicting firing from eGeMAPS (acoustic) vs sentence identity (lexical); R-squared is unusable because the pooled features are ~96% sparse.

Contributions.

(1) A confound-controlled emotion-feature definition for speech SAEs; (2) evidence that disentanglement and decodability are distinct phenomena; (3) CREMA-D cross-encoder cross-speaker generalization is sufficient characterization of the emotion-feature population.

SAE dissection of frozen-encoder emotion features (HuBERT / CREMA-D)



3. Results

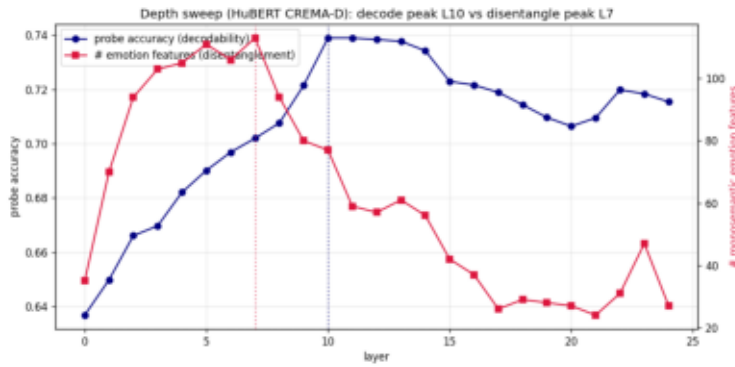


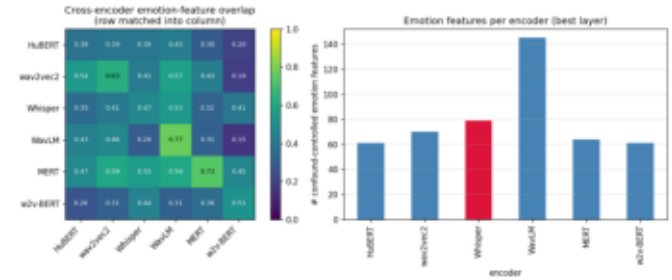
Figure 2. Depth sweep: probe accuracy (decode) peaks at L10 while the count of monosemantic emotion features peaks earlier at L7, and monosemanticity at L2. Deep layers (L17-24) retain decodability (~0.72) but lose disentanglement (24-47 features): emotion stays linearly readable while becoming distributed. Emotion organizes into clean features BEFORE the depth at which it is best decoded (Figure 2).

Q (i) Depth: decode != disentangle.

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Q (ii.1) Across pretraining paradigms.

All six encoders carry emotion features at their best layer (HuBERT 61, wav2vec2 70, Whisper 79, WavLM 145, MERT 64, w2v-BERT 61), all anger/sadness-dominated. Matching features across encoders by per-utterance firing correlation, Whisper->SSL overlap (0.40) approximately equals SSL->SSL overlap (0.39). On fixed-lexicon CREMA-D Whisper is NOT distinct: its known resistance concerns the lexical shortcut, not acoustic emotion encoding, which all encoders share (Figure 3).



Q (ii.3) Across speakers.

Under speaker-disjoint cross-validation over 91 actors, emotion accuracy is unchanged (random 0.551 vs disjoint 0.551; leave-speaker-out drop -0.000); each feature fires for ~95% of its emotion's actors. Because selection excluded speaker-entangled features, the published MESD leave-one-speaker-out collapse (~0.30) is attributable to exactly those excluded features (Figure 5).

Q (iii) Acoustic, not lexical.

Positive control (fixed lexicon => must be acoustic): 60/61 emotion features label acoustic (eGeMAPS firing-AUC 0.83 vs sentence AUC 0.66; Figure 4). Causal sufficiency: emotion decodes from the 61 acoustic features at 0.552, vs 0.431 from lexical and 0.423 from random features (6-class; full reconstruction 0.73; Figure 6). The recovered emotion signal is acoustic.

4. Discussion

Decodability and disentanglement are distinct axes. A probe reads emotion best where the representation is linearly separable (L10 and the deep layers), but the encoder packs emotion into clean monosemantic features earlier (L2-L7); deep layers trade interpretable structure for distributed separability. This sharpens the project's 'middle-to-late' decodability result into a claim about where emotion is actually organized.

On lexicon-controlled data all six encoders converge on a shared acoustic emotion basis, and those features survive speaker-disjoint evaluation. The Whisper-vs-SSL divergence reported previously is therefore not about acoustic emotion features (identical here) but about whether an encoder exploits a LEXICAL shortcut when one is available — which CREMA-D cannot pose (its text is fixed) and which the deferred EMIS experiment is designed to answer.

The causal-sufficiency result answers question (iii): emotion is decodable from the acoustic features far better than from lexical or random ones, and those features were independently validated against eGeMAPS. What these encoders expose to a probe is paralinguistic delivery, not lexical content.

5. Limitations & deferred

Utterance-pooled sparse features make necessity-ablation of small feature subsets ~null (signal is distributed across thousands of features); sufficiency is the informative test. Three planned experiments are deferred because their data is not on the cluster, each with a runnable stub: the IEMOCAP real-transcript ablation (transcripts not synced), the SpeechCraft cross-lingual arm (no audio), and the EMIS audio-vs-text feature separation (unavailable) — the decisive test of the Whisper lexical-shortcut mechanism.

6. Reproducibility

Code in `src/sae/` and `notebooks/sae/`; per-step CSVs/plots/metrics under `results/SAE/{csv,plots,json}`; combined artifacts, the summary figure and this paper under `results/SAE/summary/`. SAEs train on a single V100; `build_summary.py` and `build_paper.py` regenerate all artifacts.

References

- [1] linguistic-agnostic-ser report: emotion in middle-to-late layers; Whisper L24 resists the EMIS lexical shortcut (text-bias -0.19) while SSL encoders take it (+0.61..+0.72); MESD LOSO ~0.30; EN-ZH transfer 0.29-0.32.
- [2] Gao et al. 2024, Scaling and evaluating sparse autoencoders (TopK SAEs).
- [3] SAELens — open-source SAE training library.
- [4] arXiv:2605.12225, Sparse autoencoders for Whisper ASR (phonetic/morphemic/semantic features).
- [5] arXiv:2509.08454, Mechanistic interpretability of LoRA-adapted Whisper for SER.
- [6] eGeMAPS / openSMILE — acoustic descriptor set.

Key numbers

SAE FVU 0.070 (93% var), 4/8192 dead. Emotion features 6407 sig -> 394 mono -> 61 confound-controlled. Depth: decode L10 vs disentangle L7. Acoustic control 60/61. Sufficiency acoustic 0.552 vs lexical 0.431. Cross-speaker LOSO drop -0.000.

Figure 4. Acoustic vs lexical AUC per emotion feature (fixed lexicon). probe.

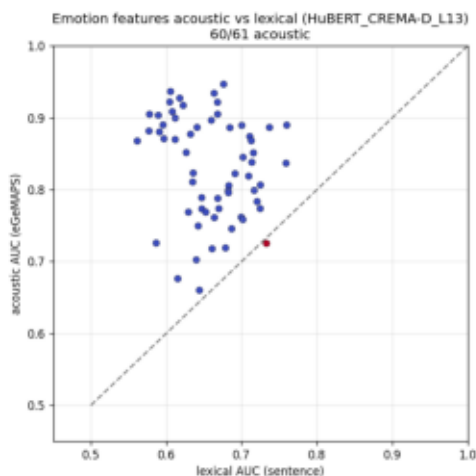


Figure 5. Cross-speaker generalization (LOSO).

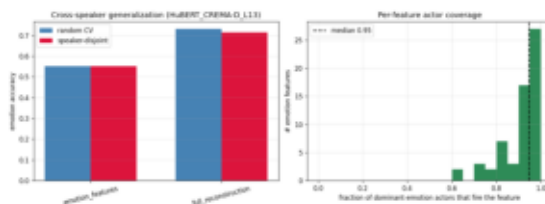


Figure 6. Causal sufficiency

